

Definition 1. Let Var be the set of propositional variables. Then ϕ is defined inductively as follows:

$$\phi ::= p \mid \phi \vee \phi \mid X\phi \mid \phi U \psi \mid \langle a \rangle \phi$$

Definition 2. Let ϕ be a formula and $p \in Var$. Then the Closure of ϕ is inductively defined as

$$\begin{aligned} Cl(p) &= Cl(\neg p) = \{p, \neg p\} \\ Cl(\neg \phi) &= Cl(\phi) \\ Cl(\phi \vee \psi) &= \{\phi \vee \psi, \neg(\phi \vee \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\ Cl(X\phi) &= \{X\phi, \neg X\phi\} \cup Cl(\phi) \\ Cl(\phi U \psi) &= \{\phi U \psi, \neg(\phi U \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\ Cl(\langle a \rangle \phi) &= \{\langle a \rangle \phi, \neg \langle a \rangle \phi\} \cup Cl(\phi) \end{aligned}$$

Definition 3. An atom A is a subset of $Cl(\phi)$ such that

$$\begin{aligned} \text{if } \psi \in Cl(\phi), & \quad \text{then } \psi \in A \text{ iff } \neg\psi \notin A, \\ \text{if } \phi_1 \vee \phi_2 \in Cl(\phi), & \quad \text{then } \phi_1 \vee \phi_2 \in A \text{ iff } \phi_1 \in A \text{ or } \phi_2 \in A \\ \text{if } \phi_1 U \phi_2 \in Cl(\phi), & \quad \text{then if } \phi_1 U \phi_2 \in A \text{ then } \psi_1 \in A \text{ or } \psi_2 \in A \\ \text{if } \phi_1 U \phi_2 \in Cl(\phi), & \quad \text{then if } \phi_2 \in A \text{ then } \phi_1 U \phi_2 \in A \end{aligned}$$

Definition 4. We say two atoms A, B are in relation $R_x(AR_x B)$ iff

$$\begin{aligned} \text{if } X\psi \in Cl(\phi), & \quad \text{then } X\psi \in A \text{ iff } \psi \in B, \\ \text{if } \psi_1 U \psi_2 \in Cl(\phi), & \quad \text{then if } \psi_1 U \psi_2, \neg\psi_2 \in A, \text{ then } \psi_1 U \psi_2 \in B, \\ & \quad \text{and if } \psi_1 U \psi_2 \in B, \psi_1 \in A, \text{ then } \psi_1 U \psi_2 \in A. \end{aligned}$$